BLACK SESAME TECHNOLOGIES

Kalos Module Datasheet

NX-3000-D

Data Sheet

Document History

Author	Version	Date	Description of Change
HaoNing	1.0	2026.1.2	Document initial
HaoNing	1.1	2026.2.6	Added : Device ordering information
HaoNing	1.2	2026.6.29	Updated and corrected technical parameters.
Shichenkang	1.3	2026.6.29	1. Converted full-width punctuation to half-width in English text; 2. removed zero-width characters.
HaoNing	1.4	2026.10.21	Update chapter 1.2 "Module Specifications Table", add the performance differences comparison between M and L modules

Sesamex

Table of Contents

1 Introduction *4*
1.1 Key Advantages 4
1.2 Module Specifications Table 4
2 Pin Definitions *7*
2.1 260 Pin List 7
3 Power Supplies *11*
3.1 Supply Current Capabilities & Interface Electrical characteristics 11
4 Mechanical Specification *12*
5 Device Ordering Information *13*

SesameX

1 Introduction

Kalos Module is a high-performance, low-power SoC designed for service robots, mobile robots, and lightweight embodied-intelligence devices. It integrates multi-core CPUs, the BSNN AI accelerator, a multi-core ISP, multiple vision interfaces, and a rich set of automotive/industrial-grade peripherals to deliver an all-in-one platform for perception, computation & control in robotics.

1.1 Key Advantages

- High-Efficiency AI:
  Equipped with the BSNN engine supporting CNN-based neural networks, ideal for object detection, segmentation, navigation, semantic understanding, and other edge AI tasks.
- Advanced Vision Fusion:
  Dual MIPI CSI interfaces with a multi-core ISP delivering 750 million pixels per second, capable of processing up to 8× 1080p@30fps streams in parallel — enabling SLAM, obstacle avoidance, and multimodal perception.
- Edge-Native Intelligent Control:
  Integrates a Cortex-A55 CPU cluster with an R5 safety MCU, balancing high-level AI workloads with low-level real-time control.
- Rich Connectivity:
  Offers PCIe, USB 3.0, and TSN-capable Ethernet, supporting expansion with additional AI accelerators, sensors, and peripherals.
- Functional Safety:
  Features dual-core lockstep MCU, AES/SHA cryptography, TRNG, OTP storage, secure boot, and fault detection — suitable for commercial robots, logistics robots, inspection robots, and similar applications.

1.2 Module Specifications Table

	NX-3000-D008-M	**NX-3000-D008-L**
AI Performance	48 TOPS INT8	16 TOPS INT8
CPU	Octa-core ARM® Cortex®-A55**(1.5GHz)** • 32KB I-and D-L1 caches in each core • 128KB combined L2 cache per core • 2MB shared L3 cache • NEON SIMD, FPU and crypto support	Octa-core ARM® Cortex®-A55**(1.2GHz)** • 32KB I-and D-L1 caches in each core • 128KB combined L2 cache per core • 2MB shared L3 cache • NEON SIMD, FPU and crypto support
GPU	• Mali-450 MP2 • Up to 218 Mtri/s Triangle-Fillrate • Up to 2 Gpix/s Pixel-Fillrate • Up to 30 Gflops	
MCU (motion control processor)	dual-core ARM Cortex-R5 @**800MHz** • DCLS(Dual-core lockstep) • Watchdog timer	dual-core ARM Cortex-R5 @**400MHz** • DCLS(Dual-core lockstep) • Watchdog timer
Memory	2 x 4GB LPDDR4 @ 4000MHz 64GB eMMC	2 x 4GB LPDDR4 @ 3200MHz 64GB eMMC

Seeed

ISP		3× ISP cores Each core runs at 400MHz, one-pixel/cycle pipeline	2× ISP cores Each core runs at 400MHz, one-pixel/cycle pipeline
		• one-pixel/cycle, each supporting up to 4 input channels and multi-view outputs. • Image processing: supports RGGB/RCCB/RYYB/4C sensors with de-mosaic, noise reduction (Raw/UV DNS), color/gamma/contrast correction, sharpening, scaling, and cropping. • Enhancements: multi-exposure HDR up to 120 dB, night-vision enhancement, LED flicker suppression. • Auto controls: adjustable AE, AWB, 33-point PWL curve, lens/dynamic color shading correction, and defect pixel compensation. • Output: 3 views per core to DDR (YUV422/420, down-scaled, or Y-channel).	
Computer Vision (DSP)		4 x DSP@800MHz	2 x DSP@800MHz
		• HW acceleration modules: GWARP (affine/perspective transform, bilinear/bicubic, LUT-based) and Scaler (up/down-scale, polyphase, pyramid). • Format support: RGB/YUV422/420 input/output across warp and scaling engines. • CV library support: OpenVX, Eigen, and standard APIs. • Core functions: stereo vision, optical flow/SfM, SLAM, image pyramid, feature extraction (HOG, Harris, ORB).	
Codec	Encode	Up to 4K@60	Up to 4*720p@25fps
		• HEVC (H.265) Main & Main10 • H.264 BP/MP/HP • VP8, VP9 (Profile 0, Profile 2 @10-bit) • JPEG (baseline, YUV420/422)	
	Decode	Up to 4K@60	Up to 4*720p@25fps
		• HEVC (H.265) Main/Main10; H.264 BP/MP/HP • VP8, VP9 (Profile 0, Profile 2 @10-bit) • VC-1 SP/MP/AP, MPEG4 SP/ASP, MPEG2 MP, H.263 P0 • RealVideo 8/9/10, JPEG (baseline, YUV420/422)	
Display		• 1x HDMI 1.4 1080p@60fps	
Networking		• Ethernet | Media Access Controller (MAC) • 2x 1Gbps • Support TSN features: • Support 802.1Qav, 802.1AS, 802.1Qbv, 802.1Qbu	
Imaging		• 2x MIPI CSI2 Rx Ports/4 data-lanes per port • Support 4 virtual channel per port,up to 8 virtual channels total • 16-bit Parallel input, up to 1080p • SLVS-EC and HiSPi sensor supported via external bridges	
External Storage		• 2KB Eeprom & 16MB NOR flash	

Sesamex

USB	• 1x USB 3.0 + 1xUSB 2.0 • Support host mode or device mode
PCIe	• Up to GEN3 | x4 (supports Root Port or Endpoint modes)
I/O	• 3xCAN FD/2xSPI/4xUART/5xI2C/2xI2S/4xPWM/ 1xJTAG
Mechanical	• Module Size: 69.6 mm x 55 mm(260pin) • 260 pin SO-DIMM Connector
Operating Requirements	• Operating Temp. Range : -25~85°C • ESD(HBM) : < 2KV

Sesamex

2 Pin Definitions

2.1 260 Pin List.

PIN (TOP)	Module Function Net Name		PIN (BOT)	Module Function Net Name
1	GND		2	GND
3	CSI1_D0_N		4	A_MIPI0_DATA2_N
5	CSI1_D0_P		6	A_MIPI0_DATA2_P
7	GND		8	GND
9	CSI1_CLK_N		10	PWR_BTN*
11	CSI1_CLK_P		12	SHUTDOWN_REQ_CTL
13	GND		14	GND
15	CSI1_D1_N		16	A_MIPI0_DATA3_N
17	CSI1_D1_P		18	A_MIPI0_DATA3_P
19	GND		20	GND
21	CSI3_D0_N		22	A_MIPI1_DATA0_N
23	CSI3_D0_P		24	A_MIPI1_DATA0_P
25	GND		26	GND
27	MCU_P3V3_S3_EN		28	A_MIPI1_CLK_N
29	GPIO236		30	A_MIPI1_CLK_P
31	GND		32	GND
33	CSI3_D1_N		34	A_MIPI1_DATA1_N
35	CSI3_D1_P		36	A_MIPI1_DATA1_P
37	GND		38	GND
39	A_ISP_FSYNC3		40	UART2_RX
41	A_ISP_FSYNC2		42	UART2_TX
43	GND		44	GND
45	A_CAMERA_PDB_3		46	UART0_RX
47	A_CAMERA_PDB_2		48	UART0_TX
49	GND		50	GND
51	A_ETH1_RXD0		52	A_ETH1_RXCLK
53	A_ETH1_RXD1		54	A_ETH1_RXCTRL
55	GND		56	GND
57	A_ETH1_RXD2		58	A_ETH1_TXD0
59	A_ETH1_RXD3		60	A_ETH1_TXD1
61	GND		62	GND
63	HDMI_TX2N		64	A_ETH1_TXD2
65	HDMI_TX2P		66	A_ETH1_TXD3
67	GND		68	GND
69	HDMI_TX1N		70	A_ETH1_GTXCLK
71	HDMI_TX1P		72	A_ETH1_TXCTRL
73	GND		74	GND
75	HDMI_TX0N		76	A_ETH1_MDIO
77	HDMI_TX0P		78	A_ETH1_MDC
79	GND		80	GND

Confidential

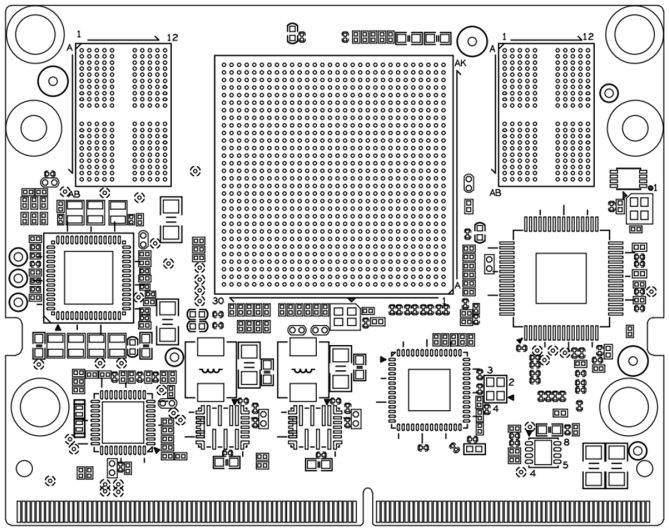
81	HDMI_TXCN	82	B_1512_RSTn
83	HDMI_TXCP	84	A_ETH1_INTR
85	GND	86	GND
87	A_USB20_VBUS	88	A_ETH0_INTR_OUT
89	SPI0_MOSI	90	A_I2C0_SCL
91	SPI0_SCK	92	A_I2C0_SDA
93	SPI0_MISO	94	HDMI_CEC
95	SPI0_CS0	96	HDMI_HPD
97	A_ERR_L1_P	98	I2C_SDA_HDMI
99	UART3_TXD	100	I2C_SCL_HDMI
101	UART3_RXD	102	GND
103	UART3_RTS	104	SPI1_MOSI
105	UART3_CTS	106	SPI1_SCK
107	GND	108	SPI1_MISO
109	USB20_DM	110	SPI1_CS0
111	USB20_DP	112	A_ERR_L1_N
113	GND	114	CAM0_PWDN
115	USB30_DM	116	CAM0_MCLK
117	USB30_DP	118	A_CAN_INT0
119	GND	120	CAM1_PWDN
121	CAN1_RX	122	CAM1_MCLK
123	CAN1_TX	124	BT_M2_WAKE_AP
125	GND	126	BT_M2_EN
127	PWR_LED_CTRL	128	W_DISABLE1_CTRL
129	GND	130	CAM_MUX_SEL
131	PCIE0_RX0_N	132	GND
133	PCIE0_RX0_P	134	PCIE0_TX0_N
135	GND	136	PCIE0_TX0_P
137	PCIE0_RX1_N	138	GND
139	PCIE0_RX1_P	140	PCIE0_TX1_N
141	GND	142	PCIE0_TX1_P
143	CAN_RX	144	GND
145	CAN_TX	146	GND
147	GND	148	PCIE1_TX0_N
149	PCIE1_RX0_N	150	PCIE1_TX0_P
151	PCIE1_RX0_P	152	GND
153	GND	154	PCIE1_TX1_N
155	PCIE1_RX1_N	156	PCIE1_TX1_P
157	PCIE1_RX1_P	158	GND
159	GND	160	PCIE2_CLK_N
161	USBSS30_RX_N	162	PCIE2_CLK_P
163	USBSS30_RX_P	164	GND
165	GND	166	USBSS30_TX_N
167	CAN2_RX	168	USBSS30_TX_P

Seeed

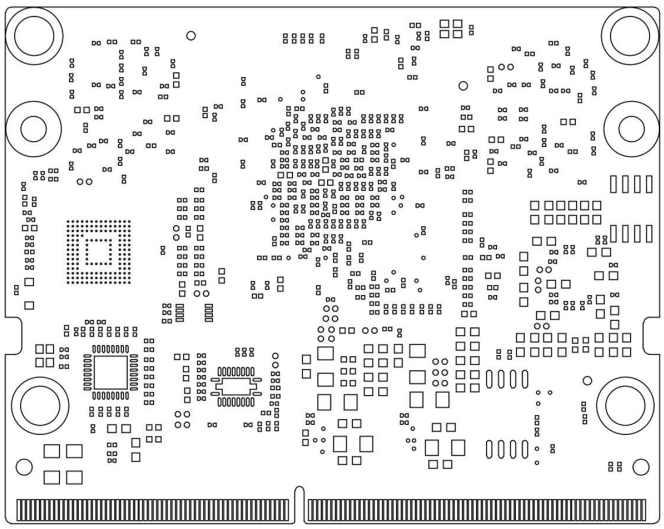
169	CAN2_TX	170	GND
171	GND	172	B_PCIE1_REF_PAD_CLK_N
173	PCIE1_CLK_N	174	B_PCIE1_REF_PAD_CLK_P
175	PCIE1_CLK_P	176	GND
177	GND	178	MOD_SLEEP*
179	PCIE_WAKE	180	PCIE0_CLKREQ
181	PCIE0_RST	182	PCIE1_CLKREQ
183	PCIE1_RST	184	GBE_MDI0_N
185	A_VOUT_SCL_3V3	186	GBE_MDI0_P
187	A_VOUT_SDA_3V3	188	GBE_LED_LINK
189	A_I2C3_SCL	190	GBE_MDI1_N
191	A_I2C3_SDA	192	GBE_MDI1_P
193	A_I2S0_SD_OUT	194	GBE_LED_ACT
195	A_GPIO_25	196	GBE_MDI2_N
197	A_I2S0_WS	198	GBE_MDI2_P
199	A_I2S0_CK	200	GND
201	GND	202	GBE_MDI3_N
203	UART1_TXD	204	GBE_MDI3_P
205	UART1_RXD	206	PWM0
207	UART1_RTS	208	FAN_TACH
209	UART1_CTS	210	A_I2S0_MCK
211	A_CAN_INT1	212	M2_ALERT*
213	CAM_I2C_SCL	214	FORCE_RECOVERY*
215	CAM_I2C_SDA	216	A_CAN_INT2
217	GND	218	PWM1
219	/	220	A_I2S1_SD_IN
221	/	222	A_I2S1_SD_OUT
223	EX_GPIO0	224	A_I2S1_WS
225	EX_GPIO1	226	A_I2S1_CK
227	A_DEBUG6	228	PWM2
229	A_DEBUG7	230	PWM3
231	GND	232	A_I2C4_SCL
233	/	234	A_I2C4_SDA
235	/	236	A_JTAG_TDO
237	POWER_EN	238	A_JTAG_TDI
239	SYS_RST*	240	/
241	GND	242	GND
243	GND	244	GND
245	GND	246	GND
247	GND	248	GND
249	GND	250	GND
251	A_VSYS_5V0	252	A_VSYS_5V0
253	A_VSYS_5V0	254	A_VSYS_5V0
255	A_VSYS_5V0	256	A_VSYS_5V0
257	A_VSYS_5V0	258	A_VSYS_5V0

Seeed

259	A_VSYS_5V0		260	A_VSYS_5V0



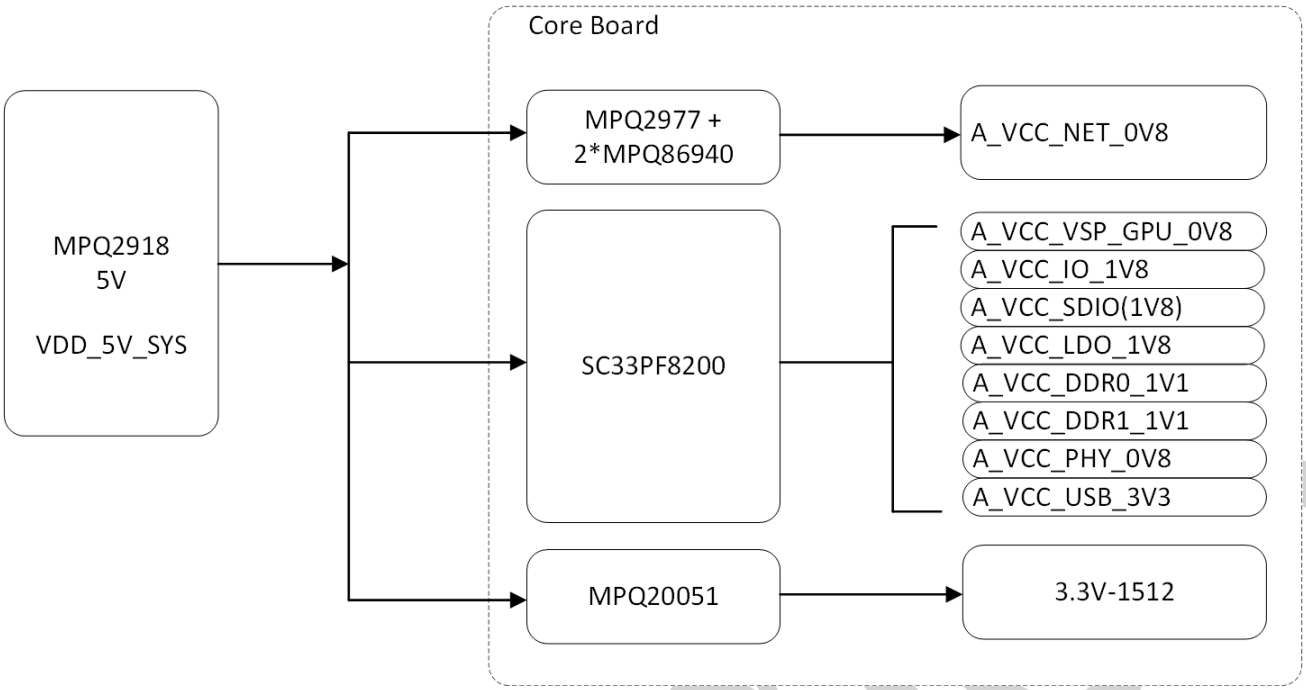
1 Top 259 Top



260 Bot 2 Bot

Sesame

3 Power Supplies



3.1 Supply Current Capabilities & Interface Electrical characteristics

Descriptions	Usage & Characteristics	Vmin	V	Vmax	Current (A)
A_VSYS_5V0	Core board 5V Power supply Input requirements	4.75	5	5.25	>7A
GPIO	Absolute input & output voltage	0	1.8	1.98	0.2
SD/EMMC IO	Absolute input & output voltage	0	3	3.3	0.2

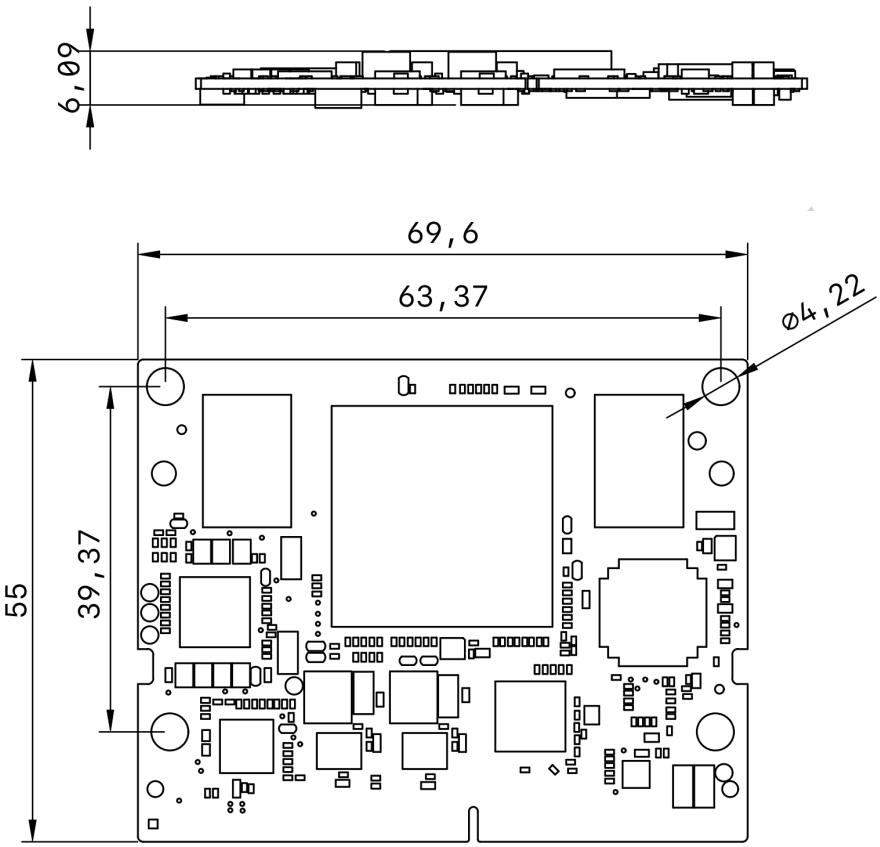
Descriptions	Min	Max
Power Consumption	15W	25W
Operational Temperature	-25℃	85℃
Operating environment humidity	10%RH	85%RH *1
Storage humidity	0	90%RH
Storage temperature	-40℃	105℃
ESD (HBM)	/	2KV

*1: 85%RH <5*24h

Seeed

4 Mechanical Specification

Unit: mm



Sesame

5 Device Ordering Information

NX-3000-D	Module Form Factor : SO-DIMM	L : Lite
NX-3000-D008-L/M	Memory Config: 8GB DRAM	M : MainStream

Sesamex